

Thalamic Connections of the Dorsomedial Visual Area in Primates

PAMELA D. BECK AND JON H. KAAS*

Department of Psychology, Vanderbilt University, Nashville, Tennessee 37240

ABSTRACT

The dorsomedial visual area (DM) of owl monkeys is a cortical area that has been described recently in a range of primate species. To study the thalamic connections of this area, injections of several distinguishable neuroanatomical tracers were placed into DM in galagos, owl monkeys, squirrel monkeys, and macaque monkeys. The distribution of label was remarkably consistent across these diverse primate species. Labeled connections were densest within the pulvinar complex. Both the lateral and inferior divisions of the pulvinar, but not the medial division, had connections with DM. Within the inferior pulvinar of monkeys, central lateral and central medial nuclei had dense connections, and the medial and posterior nuclei had sparse connections with DM. Sparser connections were revealed in the lateral geniculate nucleus and the nucleus limitans. Anterograde label was also found in the superior colliculus. The consistencies in the pattern of subcortical projections across prosimian primates, New World monkeys, and Old World monkeys support the concept that DM is a visual area common to all primates. In addition, these results provide further evidence for proposed subdivisions of the inferior pulvinar. *J. Comp. Neurol.* 396:381–398, 1998.

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The dorsomedial visual area (DM) is a subdivision of extrastriate visual cortex that was first described in New World owl monkeys (Allman and Kaas, 1975) as one of a series of postulated visual areas forming the rostral border of the second visual area, V2. Other early investigators, especially those exploring visual cortex organization in Old World macaque monkeys, did not describe a series of visual areas along the rostral border of V2 but, rather, proposed that much or all of this cortex was occupied by a single visual field, the third visual area, V3 (Cragg, 1969; Zeki, 1969). This interpretation was fortified by the results of later electrophysiological mapping studies (Gattass et al., 1988). However, differences in the connections and the response properties of neurons in the dorsal and ventral halves of the V3 region led some to consider each half as a separate visual area (Burkhalter et al., 1986). Subsequent magnetic resonance imaging studies of the organization of human visual cortex have tended to portray a single bordering area, V3 (Shipp et al., 1995), or a dorsal V3 (Serenio et al., 1995). Thus, one possibility is that the organization of extrastriate cortex varies, with New World monkeys and prosimians having a DM and Old World monkeys and humans having a V3.

In owl monkeys, DM contains a retinotopic map of both the upper and lower quadrants of the contralateral visual hemifield, is densely myelinated, and has dense connections with V1, V2, the middle temporal visual area (MT),

and part of posterior parietal cortex (Allman and Kaas, 1975; Wagor et al., 1975; Krubitzer and Kaas, 1990; Krubitzer and Kaas, 1993). Subcortical connections include the superior and inferior divisions of the pulvinar complex and the superior colliculus (Norden et al., 1978). More recently, several groups of investigators have found some of these characteristics of DM in a region of dorsomedial extrastriate visual cortex of other primates, suggesting that DM exists in these studied primates as well as in owl monkeys. Most notably, a retinotopic map of the complete contralateral hemifield in a densely myelinated oval of cortex in the expected position of DM has now been reported for New World marmosets (Rosa and Schmid, 1995) and prosimian galagos (Rosa et al., 1997). In addition, similar connection patterns between V1 and a densely myelinated DM-like region of cortex have been used as evidence for DM in galagos, marmosets, squirrel monkeys, owl monkeys, patas monkeys, and macaque monkeys (Krubitzer and Kaas, 1993). Finally, the topographic pat-

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*Correspondence to: Jon H. Kaas, Ph.D., Vanderbilt University, Department of Psychology, 301 Wilson Hall, 111 21st Avenue South, Nashville, TN 37240. E-mail: jon.kaas@vanderbilt.edu

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tern of connections of V2 with a densely myelinated DM-like field provides evidence for a retinotopically organized DM macaques (Stepniewska and Kaas, 1997). Thus, there are several lines of evidence that seem to argue against the concept of a single, large, strip-like field, V3, along most or all of the border of V2 and that favor the alternative concept that DM is a basic visual area of primates.

In part of the effort to further resolve the nature of the organization of extrastriate cortex in primates, we have placed injections of anatomical tracers in the presumptive DM region of several primate species, including owl monkeys. Here, we describe subcortical connections. The inferior pulvinar complex was of special interest, because nuclear subdivisions have been reinterpreted recently on the basis of new information on histochemical characteristics (Cusick et al., 1993; Gutierrez et al., 1995; Gutierrez and Cusick, 1997; Stepniewska and Kaas, 1997). If the DM region has similar connections across primate taxa, then this would argue that the same subdivision of visual cortex, perhaps DM, has been injected. Different patterns of connections would support the possibility that some primates have DM, whereas others have V3. Preliminary reports of these results (Beck and Kaas, 1996a) and those on the cortical connections of DM (Beck and Kaas, 1995, 1996b) have appeared elsewhere.

MATERIALS AND METHODS

Connections of the pulvinar complex and other thalamic nuclei with DM were examined by placing one to five injections of different neuroanatomical tracers into DM in four species of primates. Foci of label were localized within thalamic nuclei and in subdivisions of the pulvinar complex that were identified with various histochemical and immunocytochemical stains. DM was identified as a densely myelinated field just rostral to dorsomedial V2 and by its dense myelination.

Animals

Three galagos (*Galago garnetti*), five owl monkeys (*Aotus trivirgatus*), two squirrel monkeys (*Saimiri sciureus*), and four macaque monkeys (three *Macaca nemestrina* and one *Macaca fascicularis*) were used in this study. Three of the five owl monkey brains with DM injections were available from a related study of cortical connections (Krubitzer and Kaas, 1993). This range of species includes representatives of three major lines of primate evolution. Galagos are nocturnal, prosimian primates that represent the lorisiform branch of primate evolution. Both owl monkeys and squirrel monkeys are New World monkeys, but owl monkeys are nocturnal, and squirrel monkeys are diurnal. Macaque monkeys are diurnal Old World monkeys.

Surgery and injections

All surgical procedures were performed under sterile conditions in accordance with guidelines from the National Institutes of Health (NIH) and the Vanderbilt Animal Care Committee. Animals were initially anesthetized with ketamine hydrochloride (30 mg/kg) and xylazine (5 mg/kg), or with 3% isoflurane gas administered through a face mask. When these animals reached a surgical level of anesthesia, a dosage of 1.5–2.0% isoflurane was administered through an endotracheal tube. Dexamethasone was given pre- and

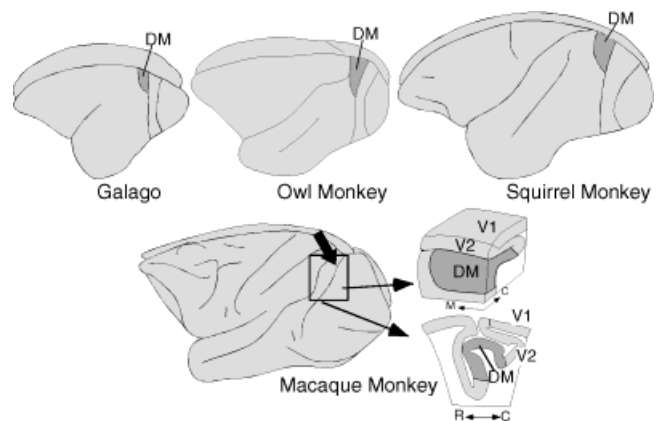


Fig. 1. The location of the dorsomedial visual area (DM) in primates. In galagos, owl monkeys, and squirrel monkeys, DM is on the dorsolateral surface of cortex. In macaque monkeys, DM is deep within the lunate sulcus (thick arrow). A three-dimensional en face view of the lunate sulcus is shown on the top right of the macaque cortex, and a parasagittal section is shown on the bottom right. V1, first visual area; V2, second visual area.

postoperatively to prevent cerebral swelling. Heart rate, temperature, and respiration were monitored throughout surgery, and Ringers' solution (subcutaneous) and antibiotics (intramuscular) were injected postoperatively.

After the head was secured in a stereotaxic frame, an incision was placed along the midline of the scalp, bone was removed, and cortex in the region of dorsomedial visual cortex was exposed. Except for macaque monkeys, the location of DM was estimated by using distances relative to sulci and the midline. Pressure injections through a glass micropipette sealed to a 1 μ l or 5 μ l Hamilton syringe (Reno, NV) were placed 0.8–1 mm below the cortical surface. In macaque monkeys, a part of the prelunate gyrus was aspirated to allow access to the caudal branch of the lunate sulcus (see Fig. 1). Injections were then placed in the cortex on the caudal, lower bank of the lunate sulcus or onto the annectent gyrus ventral to a large blood vessel that divides the upper and lower banks of the lunate sulcus. Typically, different neuroanatomical tracers were placed at several sites within the expected location of DM. Injected tracers included wheat germ agglutinin conjugated with horseradish peroxidase (WGA-HRP; 0.02 μ l of 2% WGA-HRP in saline), Fast blue (FB; 0.1–0.2 μ l of 2% FB in saline), Fluororuby (FR; 0.1–0.3 μ l of 10% FR in saline), Diamidino yellow (DY; 0.2–0.3 μ l of 2% DY in dH₂O), Fluorescein (FS; 0.2 μ l of 2% FS in saline), and Fluoroemerald (FE; 0.2 μ l of 10% FE in saline).

After the injections were completed, the exposed cortex was covered with dura, a sterile, gas-permeable contact lens, and Gelfoam. In addition, a small piece of Gelfoam was placed in the lesion created by aspiration in macaque monkeys. The craniotomy was covered by a cap of quick-curing dental cement that was secured to the cranium with the same material. The wound was sutured, and anesthesia was discontinued and replaced with oxygen until the animal showed signs of recovery. The endotracheal tube was then removed, and the animal was allowed to recover fully from anesthesia on a warming pad.

Histology

Following a survival period of 7–14 days, sufficient for transport of the tracers, animals were initially anesthetized with ketamine hydrochloride and xylazine and were then given a lethal dose of sodium pentobarbital. When they were areflexive, animals were perfused with physiological saline, followed by 2% paraformaldehyde in phosphate buffer, then 2% paraformaldehyde with 10% sucrose in phosphate buffer (0.1M), pH 7.4. The cranium was opened, and the brain was removed. Cortex and thalamus were separated, and the cortex was processed for a series of related studies on cortical connections. The thalamus was stored overnight in a solution of 30% sucrose in phosphate buffer.

The following day, the thalamus was cut and processed for histology. Sections were cut 50 μ m thick in a coronal plane on a freezing microtome and reserved in phosphate buffer. Every fourth to fifth section was treated for either tetramethylbenzidine (TMB; Mesulam, 1978), cytochrome oxidase (CO; Wong-Riley, 1979), acetylcholinesterase (AChE; Geneser-Jensen and Blackstad, 1971), or calbindin (CB; Celio, 1990) or was left unprocessed for fluorescence microscopy. Sections were then mounted onto gelatin-coated glass slides, allowed to dry, and coverslipped.

Data analysis

Brain sections were drawn showing locations of labeled cells, terminals, and architectonic borders. The locations of fluorescent labeling were digitized by using a fluorescent microscope equipped with a computer-interfaced X-Y encoder and were then printed at a low magnification by using the Bioquant image-analysis system (R and M Biometrics, Nashville, TN). Label from the WGA-HRP injection and borders of thalamic nuclei were reconstructed at the same magnification by using a drawing tube attached to a lightfield/darkfield microscope.

RESULTS

In this report, we describe the subcortical projections of the visual area DM (or the presumptive DM) in four taxonomic groups of primates. The locations and borders of DM shown for galagos, owl monkeys, squirrel monkeys, and macaque monkeys in Figure 1 were determined in the present study from the myeloarchitecture of dorsomedial visual cortex studied in brain sections cut parallel to the surface of manually flattened cortex. In all of these primates, DM was defined as a darkly myelinated region along the rostral border of V2. The myeloarchitectonic borders of DM in owl monkeys, squirrel monkeys, galagos, macaques, and other monkeys have also been demonstrated repeatedly elsewhere (Allman and Kaas, 1975; Wagor et al., 1975; Krubitzer and Kaas, 1990, 1993; Rosa and Schmid, 1995; Stepniewska and Kaas, 1996). In the present cases, the border with V2 was most obvious due to the characteristic banding pattern of V2 (Livingstone and Hubel, 1982; Stepniewska and Kaas, 1997). The precise locations of other borders were less certain, but the general extent of DM was clear. Histological documentation on the proposed borders of DM will be presented elsewhere with our reports on cortical connections. The location of DM in the present cases closely corresponds to the region defined as DM in previous studies (see Discussion). All injections

judged to be within DM in the present cases labeled neurons in V1 and V2, structures that are known to project to DM (Krubitzer and Kaas, 1993; Stepniewska and Kaas, 1996). Here, we describe the subcortical connections of DM with the pulvinar complex, the dorsal lateral geniculate nucleus (LGN), the nucleus limitans, and the superior colliculus. Other subcortical connections with the thalamic reticular nucleus, the claustrum, and the caudate nucleus were observed, but they are not described here.

Galagos

The subcortical connections were studied after a total of eight distinguishable injections or sets of injections into DM of three galagos. In one animal, injections of four different fluorescent tracers were made into the caudal half of DM (Fig. 3). In the second animal, one large injection of WGA-HRP was placed in central-medial DM, and a smaller and more restricted DY injection was placed more caudally (Figs. 4, 5). In the third animal, one large injection of FB was placed in central-medial DM, and a smaller DY injection was made more caudally (not shown). Diffusion from the WGA-HRP injection of the second animal (case 95–09) involved cortex rostral to DM, but labeled connections were similar to those from injections that were completely within DM, so any transport from outside DM was minimal.

Much of the subcortical label was in the pulvinar complex, which is not highly differentiated in galagos, but the three major divisions that are typical of primates can be distinguished (Fig. 2). The inferior pulvinar (PI), which is a complex of at least four nuclei in monkeys (Stepniewska and Kaas, 1997), is distinguished by its location between the LGN and the brachium of the superior colliculus. Clear subdivisions of the inferior pulvinar in galagos were not apparent in our AChE, CB, and CO preparations, even though these preparations allowed four nuclei of the inferior pulvinar to be delimited in monkeys (see below). The lateral (PL) and medial (PM) divisions of the pulvinar were not well differentiated, and they have been commonly grouped together as the superior pulvinar in previous studies (Symonds and Kaas, 1978; Raczkowski and Diamond, 1981). Yet, a border between the two divisions can be approximated in sections processed for CB, because PM is somewhat darker (Fig. 2B). In AChE- and CO-stained sections, PL can be somewhat darker. In the LGN, as others have reported (Casagrande, 1994), the koniocellular layers are dark in CB preparation and light in CO and AChE preparations.

Most of the subcortically labeled neurons were in the lateral and inferior divisions of the pulvinar complex (Figs. 3, 4). At least caudally, the territory of the medial pulvinar did not contain labeled neurons, and it is possible that the medial extent of labeled neurons represented the extent of the lateral pulvinar, with few or no labeled neurons in the medial pulvinar. In both the lateral and inferior pulvinar, foci of labeled neurons were scattered and highly overlapping for injections of different tracers placed within 2 mm of one another, suggesting an overlap of cortical arbors from pulvinar neurons of 1–2 mm. Yet, there was a tendency for more rostral injections into DM to label more dorsal parts of the lateral pulvinar (see, e.g., Fig. 4, section 136). In addition, the clustering of labeled neurons medially and laterally in the lateral pulvinar, especially rostrally in case 95–09 (Fig. 4), suggests the possibility of

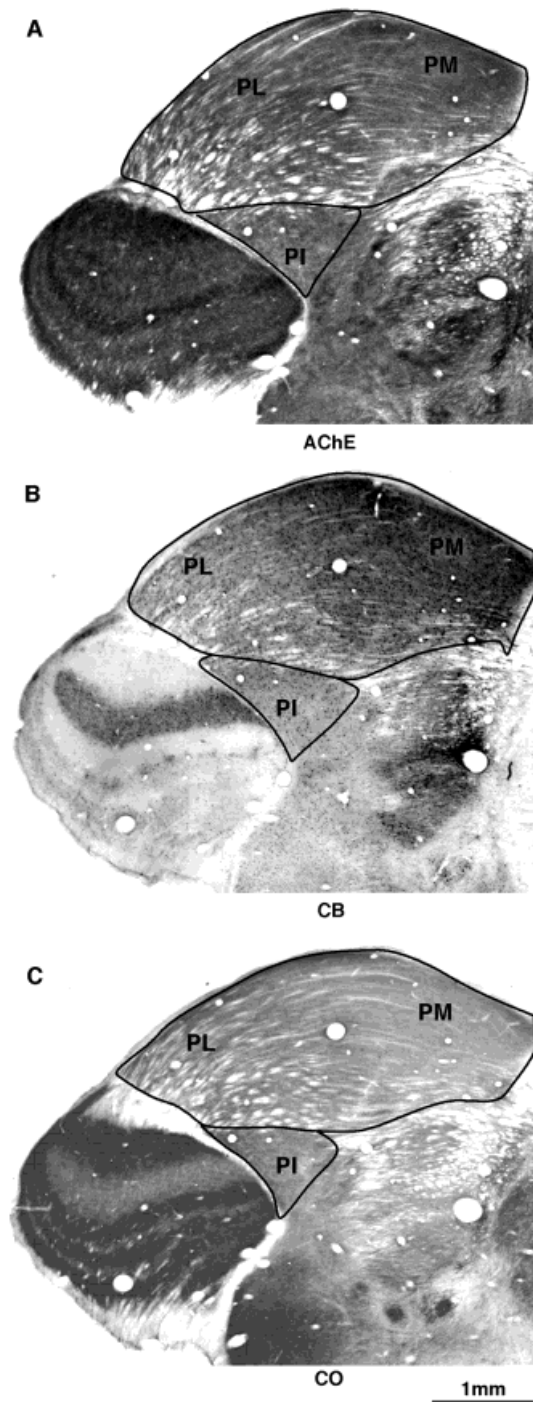


Fig. 2. **A–C:** Architecture of the pulvinar in galagos. Lateral pulvinar (PL), medial pulvinar (PM), and inferior pulvinar (PI) could not be distinguished reliably on the basis of architecture alone, but, in some sections, PM stained darker for calbindin (B). The locations of PM and PL were estimated from locations reported by Symonds and Kaas (1978), and PL always had large fiber bundles coursing laterally. PI was identified as the region under the brachium of the superior colliculus. AChE, acetylcholinesterase; CB, calbindin; CO, cytochrome oxidase.

separate nuclei. Separate nuclei may also exist in the inferior pulvinar, but the evidence from connection patterns is not compelling. Anterograde label in the pulvinar

following the WGA-HRP injection was located in the same regions as labeled cells (see Fig. 4).

Subcortically, moderate densities of labeled neurons were located in structures other than the pulvinar complex. Labeled neurons were found in the LGN following five of the eight injections into DM and were located in parvo-, magno-, and koniocellular layers. Labeled neurons in the limitans were dense following six of the eight injections (see, e.g., Fig. 3, sections 94–110). Following an injection of WGA-HRP, labeled terminals were located in the most ventral superficial layer (SO) and in the intermediate layer (SGI) of the medial half of the superior colliculus, where the lower visual field is represented. Anterograde labeling was also located in the claustrum (not shown) and the substantia nigra (Fig. 5).

Owl monkeys

Descriptions of subcortical connections in owl monkeys are based on a total of seven injections placed in DM of five animals, three of which were available from a previous study on cortical connections (Krubitzer and Kaas, 1993). Results for two of these cases (94–3 and 93–36) are illustrated here (Figs. 6, 7). In one animal, a small DY injection was placed in caudal-lateral DM, and three larger injections of FR, FB, and WGA-HRP were made into rostral-medial DM (Fig. 6). In a second animal, a large WGA-HRP injection was placed in central-rostral DM and also involved cortex rostral to DM in the ventral posterior parietal area (Fig. 7). One large injection of WGA-HRP, placed centrally in dorsal DM, was placed in each of three additional owl monkeys.

Subdivisions of the pulvinar in owl monkeys could be distinguished on the basis of staining for CB, CO, and AChE. The lateral pulvinar could be distinguished as a large nucleus above the brachium of the superior colliculus that stained somewhat darker than the medial division in CB preparations. Subdivisions of the inferior pulvinar were highly differentiated and were most obvious in sections stained for CB (for owl monkeys, see Stepniewska and Kaas, 1997; for squirrel and macaque monkeys, see Fig. 8). A moderately stained PI_{CL} was located most laterally, and, more medially, PI_{CM} stained darkly for CB, which contrasted with the light staining in PI_M . Posterior-medial PI_P showed moderate-to-dense staining for CB. Divisions were also apparent in CO and AChE preparations (see Stepniewska and Kaas, 1997). The inferior pulvinar sometimes extended slightly dorsal to the brachium of the superior colliculus, as reported by others (Stepniewska and Kaas, 1997).

Most of the labeled neurons were in the lateral and inferior divisions of the pulvinar complex. In PL, labeled neurons extended to the border with PM, but no labeled neurons were found in PM. In the inferior pulvinar, the densest label was within PI_{CL} and PI_{CM} . Seven injections that were confined to DM produced labeled neurons rarely in PI_M and never in PI_P (see, e.g., Fig. 6). The one injection that involved ventral posterior parietal cortex (VPP) labeled many neurons in PI_M and PI_P as well as in PI_{CL} and PI_{CM} (Fig. 7). Overall, labeled neurons were scattered over a wide extent of PL, PI_{CL} , and PI_{CM} , and labeled neurons from different injections were highly overlapping. However, labeled neurons from the caudolateral injection shown in Figure 6 were located medially in caudal PL and laterally in rostral PL compared with label from the more rostromedial injections. Separate clusters of labeled neu-

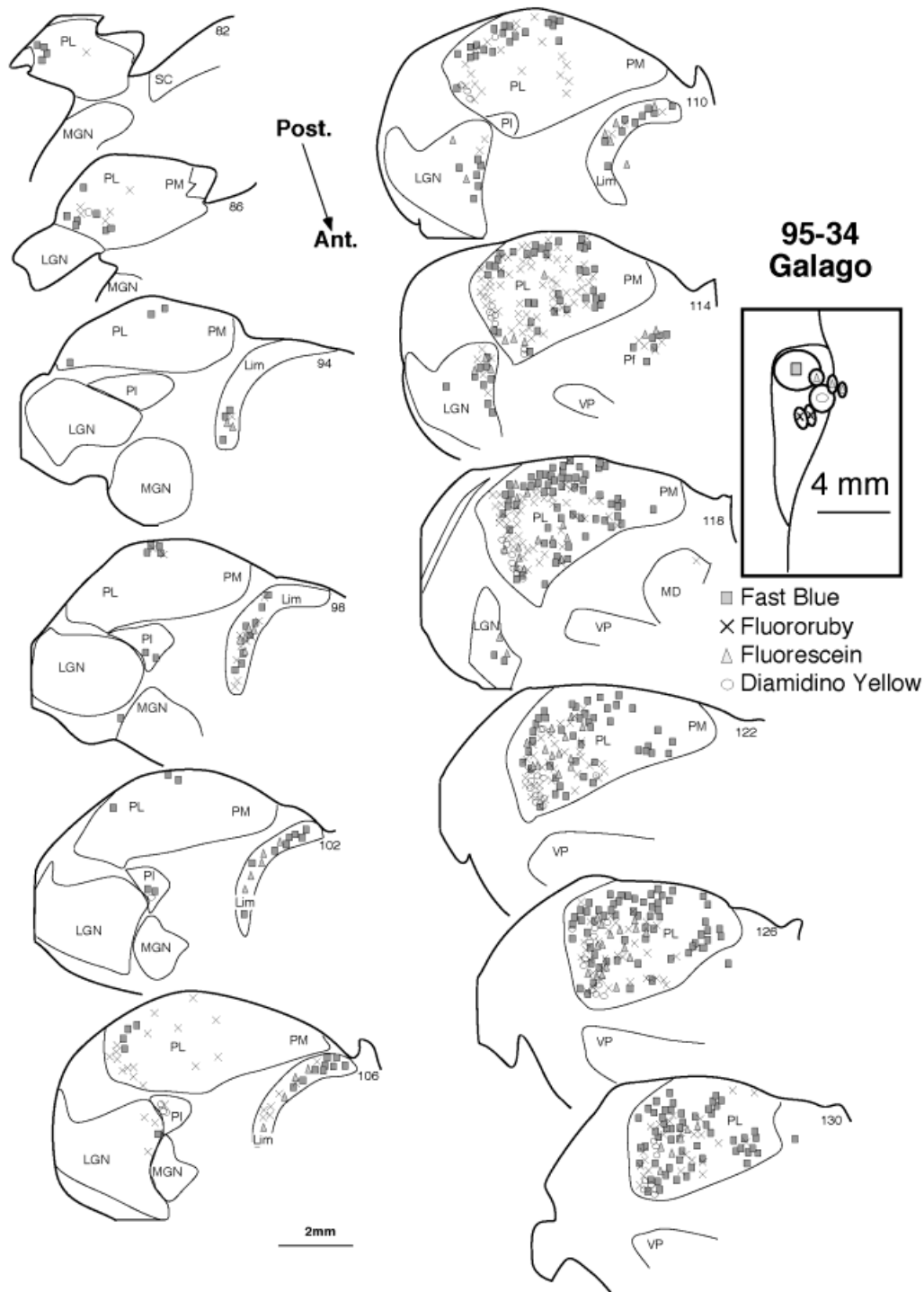


Fig. 3. Label from injections of four tracers in galago case 95-34. The location of the injection sites is shown in the box on the right. Section numbers are indicated in the upper right of each section, with the lower number corresponding to more caudal sections. LGN, lateral geniculate nucleus; Lim, limbans; MD, medial dorsal nucleus; MGN, medial geniculate nucleus; SC, superior colliculus; VP, ventroposterior nucleus.

rons in PL suggested the possibility of at least two separate nuclei.

Although most of the subcortically labeled neurons were located in the pulvinar complex, other structures contained a few labeled neurons as well. The LGN contained a

few labeled cells in the internal and external magnocellular layers following two of the eight injections. Labeled neurons and terminals were located in the limbans following one injection and in the claustrum following two injections. The injection of WGA-HRP into DM and VPP produced a few

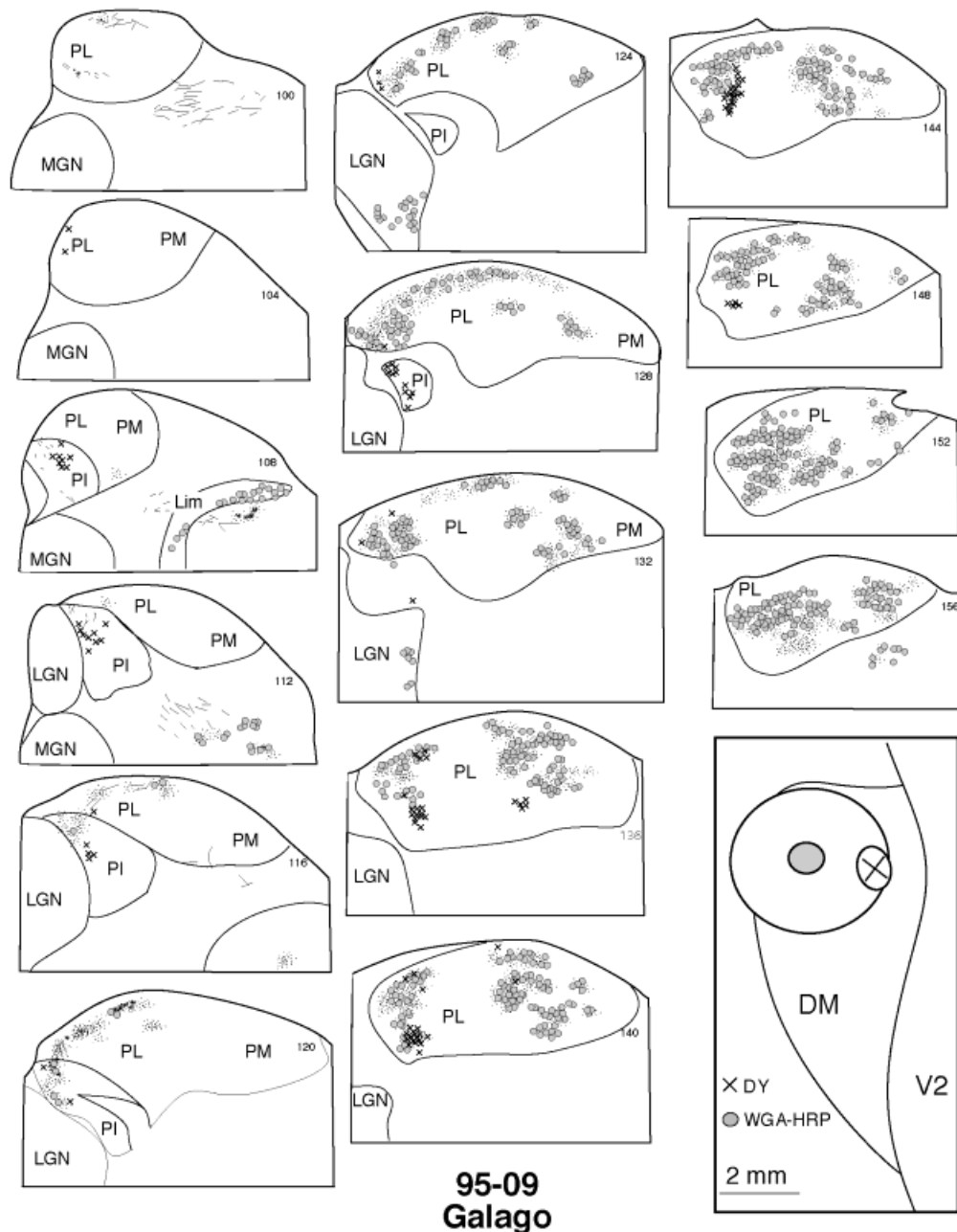


Fig. 4. Label from two injections in galago case 95-09. Retrograde label from the wheat germ agglutinin conjugated to horseradish peroxidase (WGA-HRP) injection is shown with gray circles, and anterograde is shown with stippling. Retrograde label from the Diamidino yellow (DY) injection is indicated by crosses. For other conventions, see Figure 3.

labeled neurons that were located in the ventroposterior nucleus proper and in the inferior division of the ventroposterior nucleus (sections not shown in Fig. 7). These neurons most likely were labeled by the involvement of VPP.

Squirrel monkeys

Subcortical connections in squirrel monkeys were evaluated following eight injections into DM of two animals, one of which had injections of FR and DY placed caudally and FB and FS placed rostrally in central DM (Fig. 9). In the second squirrel monkey, a small DY injection was placed

caudally, a large FB injection was placed centrally, and a small FS injection and two small FR injections were placed rostrally in DM (Fig. 10).

Many of the labeled neurons were located in the pulvinar, which can be divided into at least six subnuclei by using CO, AChE, and especially CB preparations (Stepniewska and Kaas, 1997). Briefly, PL was darker than PM in CB and AChE preparations but lighter in CO preparations (Fig. 8A-C). The inferior pulvinar had four highly differentiated nuclei, with PI_{CL} staining moderately in each preparation, PI_{CM} staining intensely for CB, PI_M

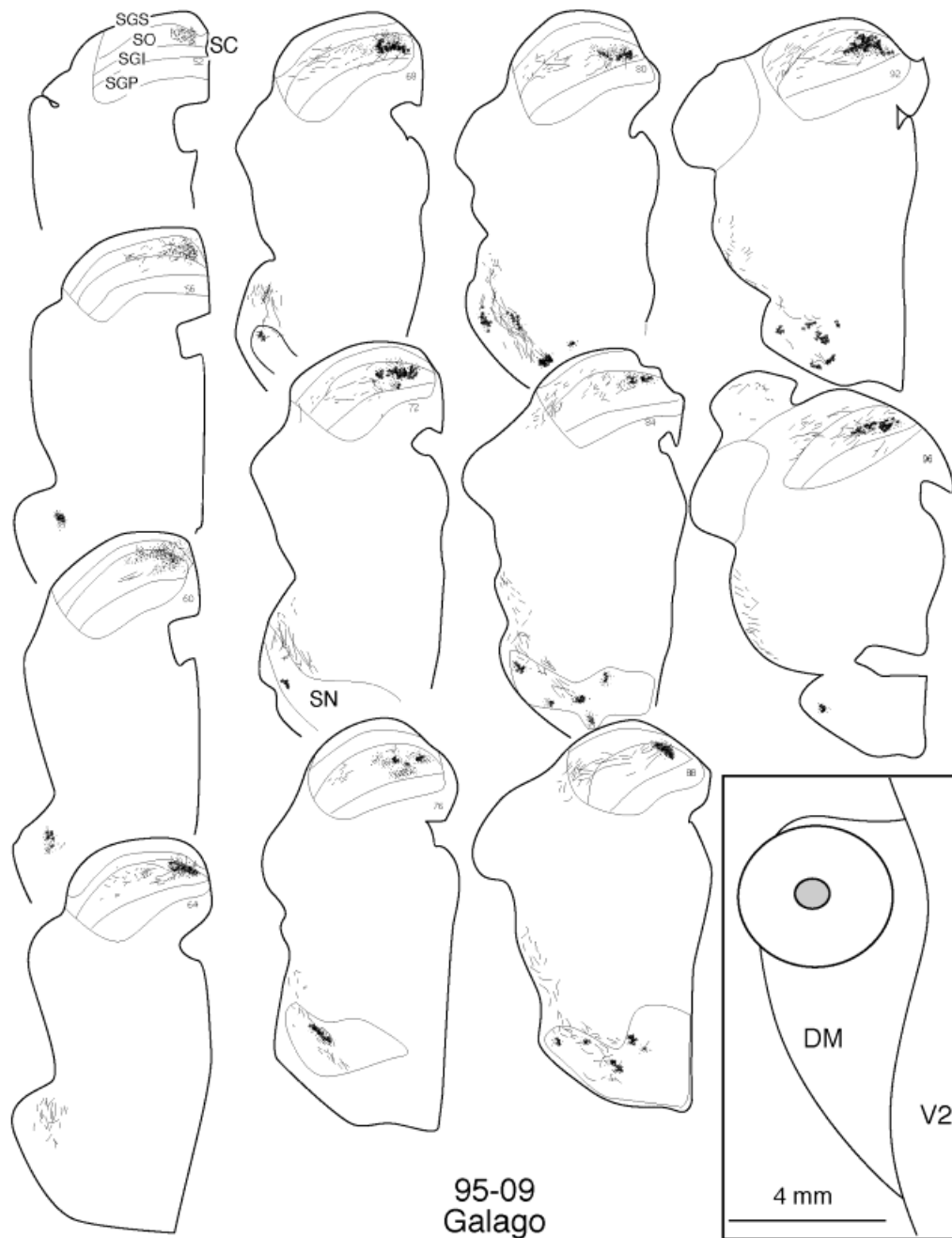


Fig. 5. Anterograde label in the superior colliculus (SC), substantia nigra (SN), and pons in galago case 95-09 following a large injection of wheat germ agglutinin conjugated to horseradish peroxidase (WGA-HRP). For conventions, see Figure 3. Layers of the SC

include the stratum griseum superficiale (SGS), the stratum opticum (SO), the stratum griseum intermediale (SGI), and the stratum griseum profundum (SGP).

staining very lightly for CB and containing dark CO and AChE patches, and PI_P staining moderately dark to dark in each preparation.

Subcortically labeled neurons in the pulvinar were located in the lateral and inferior divisions, and no labeled neurons were found in the medial pulvinar. In the lateral pulvinar, labeled neurons were located in a dorsal-to-ventral band that was confined to the lateral half of PL (see, e.g., Fig. 9, section 88; Fig. 10, section 92). More rostrally, labeled neurons were scattered over a wider

mediolateral extent, suggesting possible lateral and medial subdivisions of PL. Despite the separation of injections into caudal and rostral halves of DM, the territories of labeled neurons from different injections overlapped, suggesting thalamocortical projections that diverge up to 4 mm. In the inferior pulvinar, much of the label was in PI_{CL} and PI_{CM} , and distributions of labeled neurons from the caudal injections were centered dorsal to those from the more rostral injections. In addition, neurons labeled from the rostral injections were located caudal to neurons

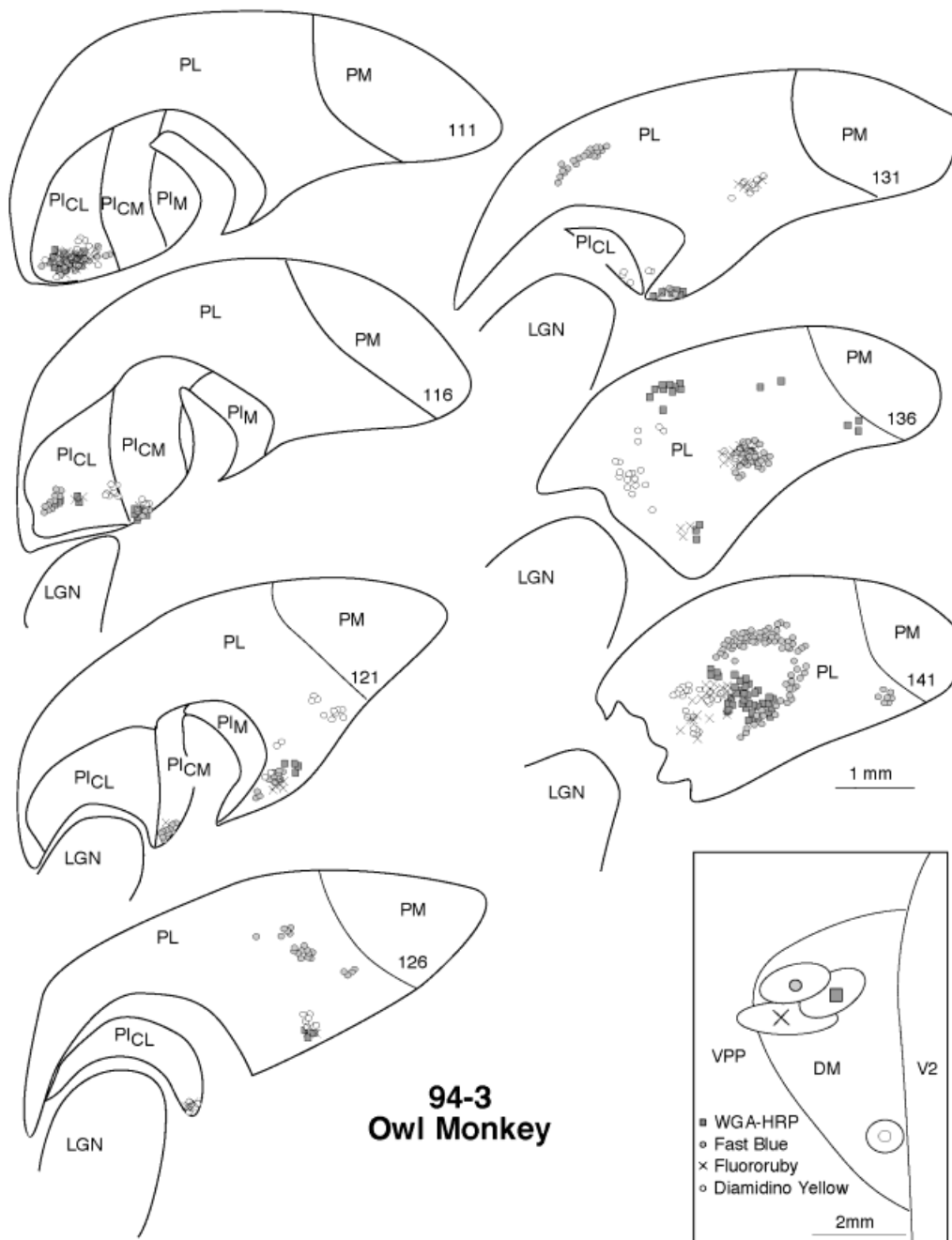


Fig. 6. Four injections and resulting label in owl monkey case 94-3. The locations of the injection sites are shown on the bottom right. The central lateral (PI_{CL}), central medial (PI_{CM}), and medial (PI_M) divisions of the inferior pulvinar are indicated. VPP, ventral posterior parietal cortex. For other conventions, see Figure 3.

labeled from caudal injections. This topological distribution is most obvious in the DY and FB label in sections 96 and 100 of Figure 9. A few labeled neurons were found in PI_M and PI_P.

A few labeled neurons were found in structures other than the pulvinar complex. In each animal, two of the four injections produced labeled neurons in the LGN (Fig. 9, sections 100 and 108; Fig. 10, sections 97 and 102). These

cells tended to be large neurons and were located in the magnocellular layers. A few smaller neurons were located in the interlaminar zone between the external magnocellular layer and the inferior pulvinar. Labeled neurons from two of the injections in one animal were located in the nucleus limitans (Fig. 9, section 112), with label from the most caudal injection positioned dorsal to label from the rostral injection.

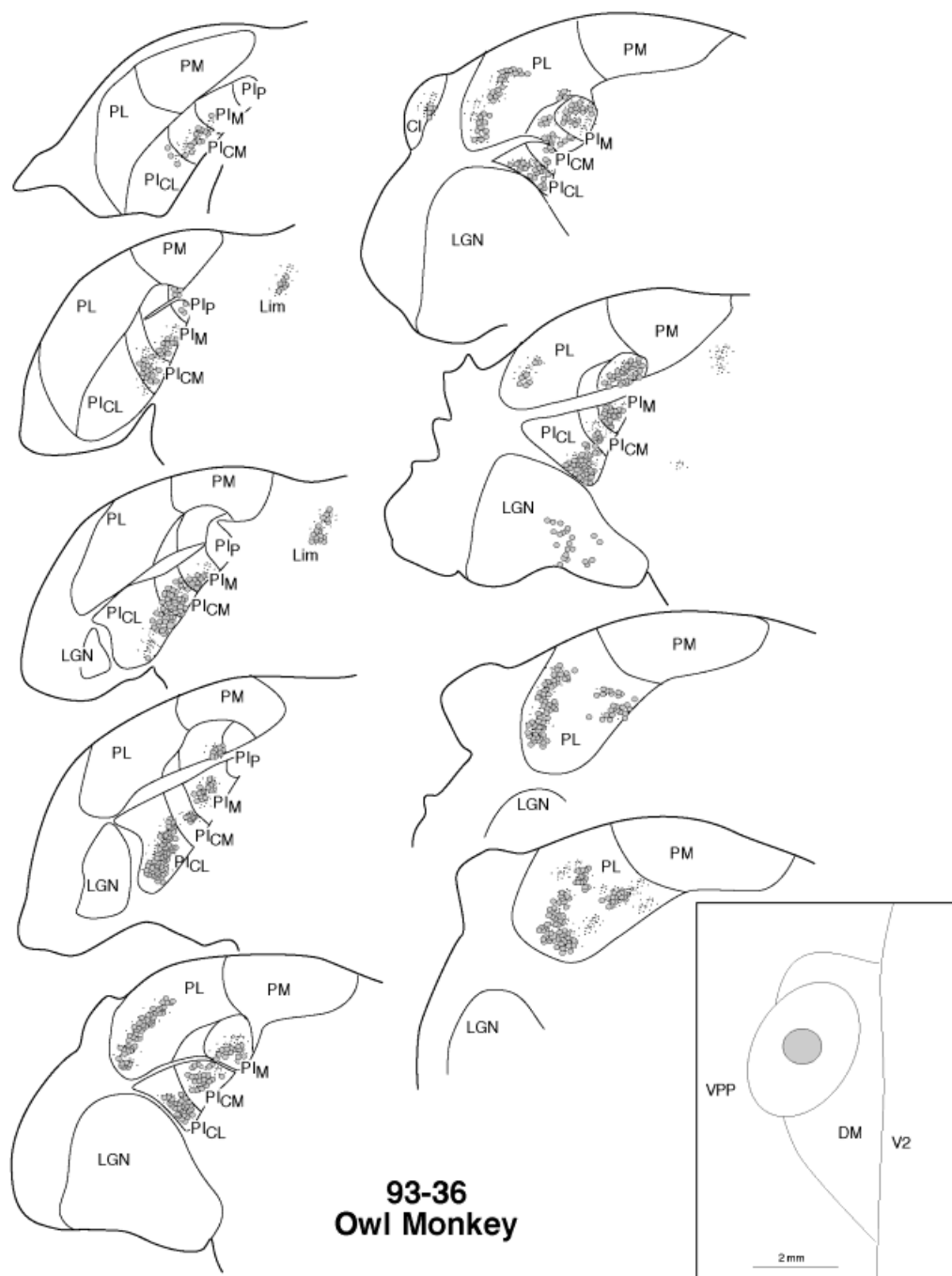


Fig. 7. One injection of wheat germ agglutinin conjugated to horseradish peroxidase (WGA-HRP) and resulting retrograde and anterograde label in owl monkey case 93-36. PI_p, posterior divisions of the inferior pulvinar. For other conventions, see Figure 6.

Macaque monkeys

Subcortical connections in macaque monkeys were determined from a total of 12 injections in four animals. Results from two of these cases are illustrated here, one with five injections of fluorescent tracers and WGA-HRP placed caudally in central DM (Fig. 11) and one with injections of DY in the central-medial part, FR in the central-rostral part, and FB midway between the other two (Fig. 12).

Photomicrographs of PI neurons and terminals labeled from a WGA-HRP injection and thalamic architecture in a third animal are shown in Figure 13. Three injections of fluorescent tracers were placed centrally in DM in the fourth macaque monkey (not shown).

Divisions of the pulvinar complex were well differentiated and easily distinguishable in CB, CO, and AChE preparations (Fig. 8D-F; see also Stepniewska and Kaas, 1997). Briefly, architectonic subdivisions of the pulvinar

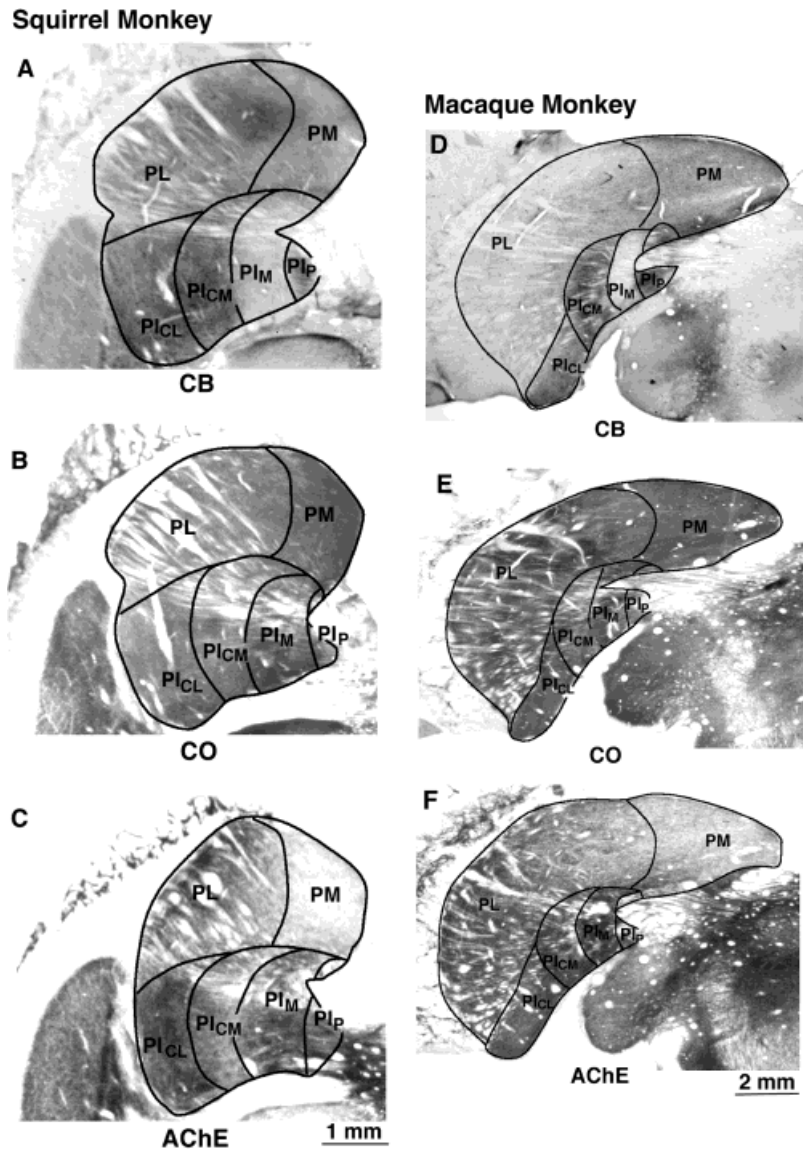


Fig. 8. Architecture of the pulvinar in squirrel monkeys (A–C) and macaque monkeys (D–F). Divisions based on calbindin (CB; A,D), cytochrome oxidase (CO; B,E), and acetylcholinesterase (AChE; C,F) were more apparent in squirrel monkeys than in owl monkeys or galagos. Medial pulvinar (PM) is apparent in AChE preparations, in which it stains lighter than adjacent lateral pulvinar (PL). PL contained large fiber bundles, and, in CO and AChE preparations, the lateral portion stained darker than the medial portion. Four divisions of the inferior pulvinar were apparent with all three stains. In CB

sections, PI_{CM} was the most intensely staining division, PI_P and PI_{CL} stained moderately, and PI_M was characterized by very light staining. In CO sections, PI_M was the darkest division, PI_{CM} stained the lightest, and PI_P and PI_{CL} stained moderately. In AChE sections, PI_{CL} stained moderately to darkly, PI_M and PI_P stained moderately, PI_{CM} was the lightest subdivision, and the lateral half was slightly darker than the medial half. In AChE and CO preparations, PI_{CM} and PI_M contained dark-staining patches.

complex were similar in appearance to those in squirrel monkeys and were distinguished most easily in CB preparations. PI_{CM} was identified easily by intense staining for CB, and PI_M was identified easily by very light staining for CB (Fig. 13B).

Much of the subcortical label was located in the lateral and inferior nuclei of the pulvinar complex. Labeled neurons in PL were widely scattered but formed two clusters in each case. A lateral cluster extended dorsoventrally in PL (see Fig. 11, section 67; Fig. 12, section 108), whereas a second cluster was located medially near the border with PM (see Fig. 11, section 87; Fig. 12, section 116). In three of

the animals, neurons labeled from the separate injections overlapped in PL, so that the locations of labeled neurons were not distinctly different for different injection sites. Overlap in the distributions of labeled neurons from injections up to 2 mm apart suggested a divergent projection from PL to DM, with thalamocortical axons invading at least 2 mm of cortical territory. When the injections extended across 6.5 mm of cortex (Fig. 12), the locations of labeled neurons in PL differed more clearly with the location of the injection into DM. A topological order of connections was most apparent in the more caudal sections (Fig. 12, sections 100–112), in which the labeled

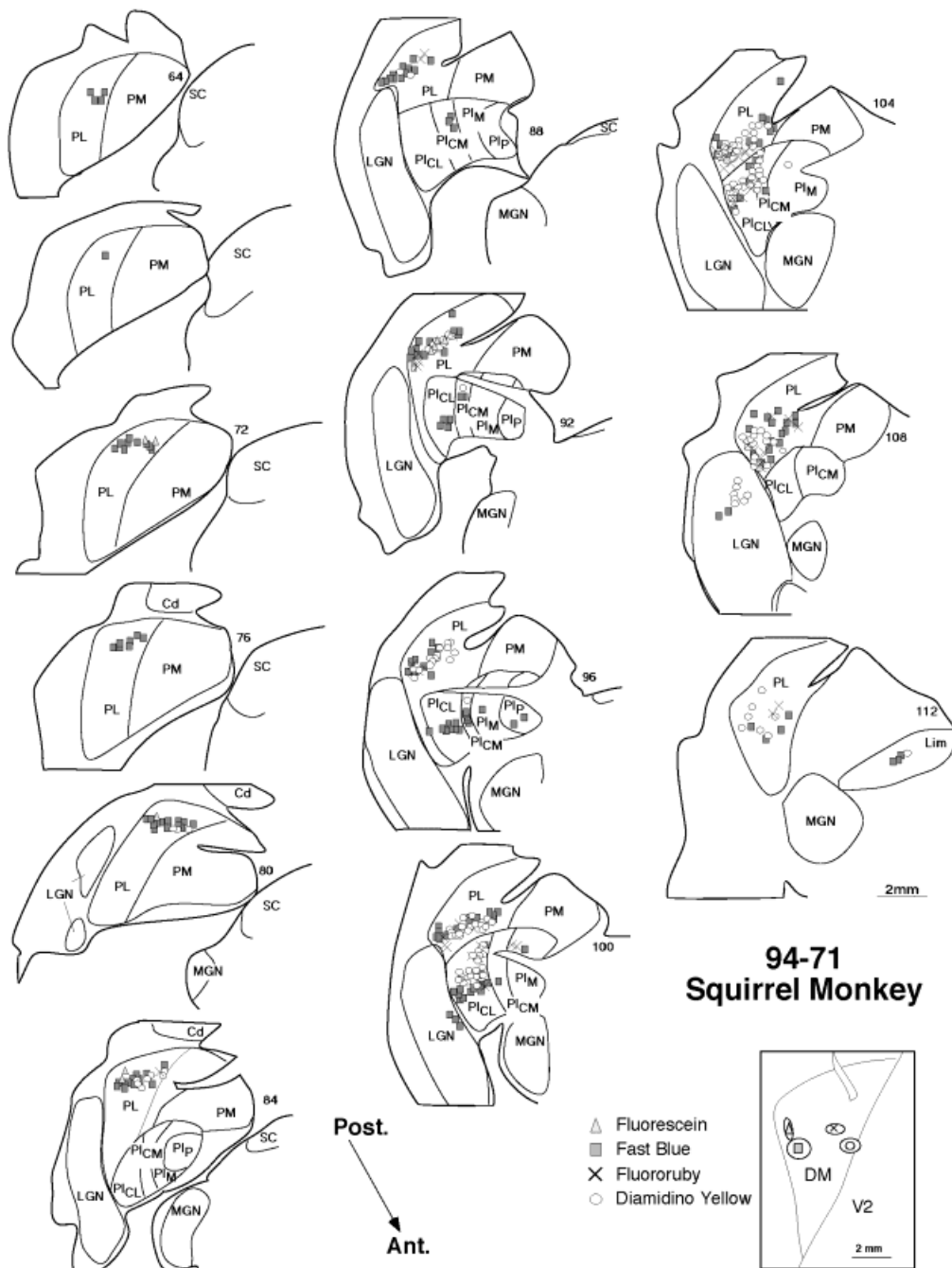


Fig. 9. Four injections and resulting label in squirrel monkey case 94-71. Cd, caudate nucleus. For conventions, see Figure 6.

neurons from the medial DY injection were positioned lateral and ventral to the labeled neurons from the lateral FR injection. Most of the labeled neurons in the inferior pulvinar were located in PI_{CL} and PI_{CM} , but three injections in two animals also produced a few labeled neurons in PI_M . No labeled neurons were found in PI_P . In Figure 12, the two injections that produced labeled neurons in PI_M were the most rostral injections; thus, the label in PI_M could be due to the injections involving cortex rostral to

DM. However, an injection well within the borders of DM in a separate case also produced label in PI_M , and DM injections in other species sometimes labeled neurons in PI_M . Thus, the PI_M label in Figure 12 likely reflects connections with DM and not surrounding cortical areas. Labeled neurons in the inferior pulvinar from different injections were largely overlapping, but label from the FR injection placed more rostral in DM tended to be located lateral to the other label in PI_{CL} and caudal in PI_{CM} (see,

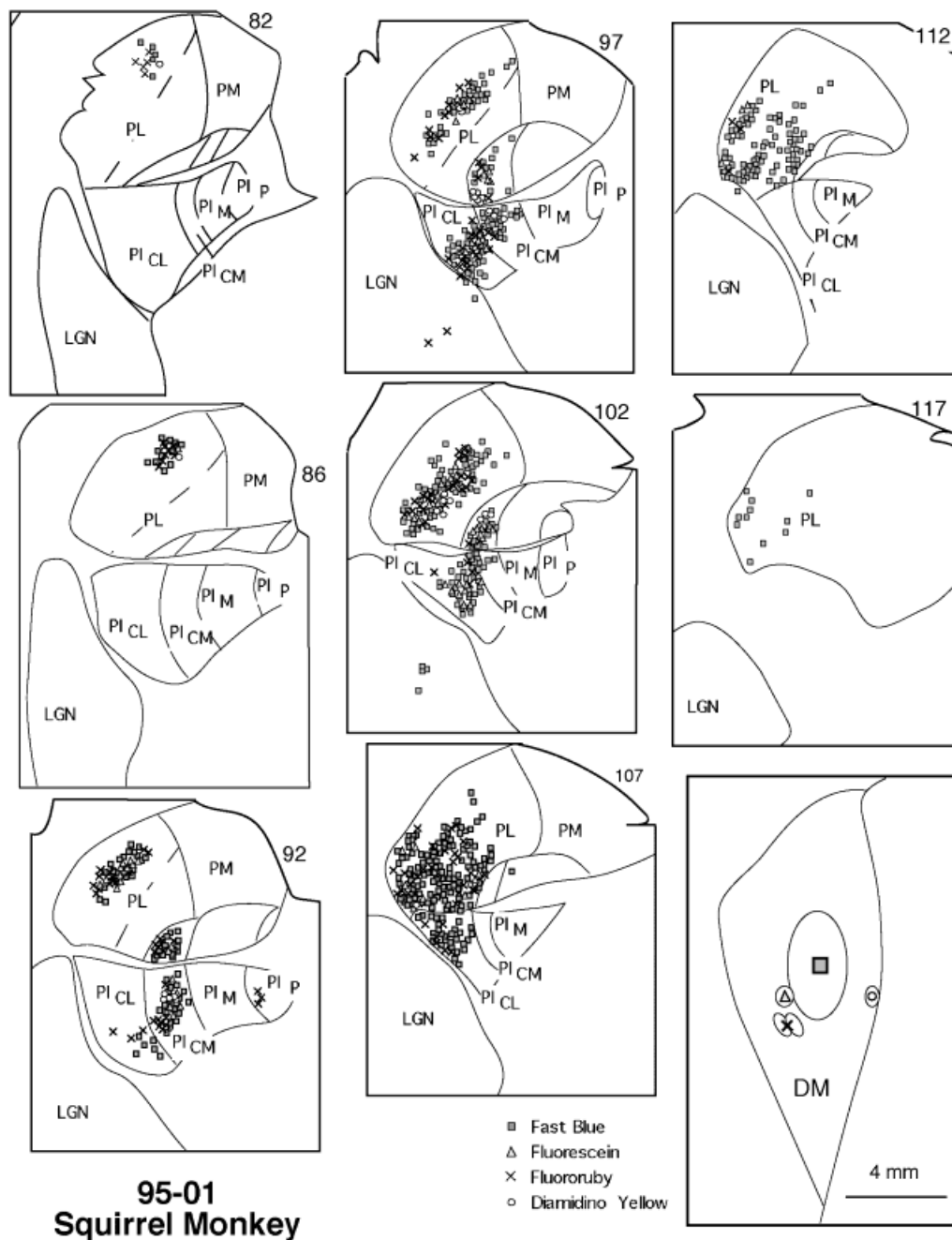


Fig. 10. Injections of four tracers and resulting label in squirrel monkey case 95-01. For conventions, see Figure 6.

Fig. 12, sections 112-124). Anterogradely labeled terminals were coextensive with labeled neurons in PL, PI_{CL}, and PI_{CM} following a WGA-HRP injection (not shown).

A few labeled neurons were found subcortically in structures other than the pulvinar complex. Following 5 of the 11 injections, labeled neurons were found in the magnocellular layers of the LGN. Three of these injections produced label in the internal magnocellular layer, and two produced label in the external layer. Labeled neurons were found in the limitans nucleus after only one of the injections (Fig. 12, section 136). Labeled terminals from the

WGA-HRP injection were found in the ventral superficial layers (SO) and the intermediate layer (SGI) of the superior colliculus. Dense patches of labeled terminals were found in the claustrum.

DISCUSSION

The major finding of the present study is that the region of extrastriate visual cortex that we define by location and myeloarchitecture as DM has similar subcortical connections in prosimian galagos, New World owl and squirrel

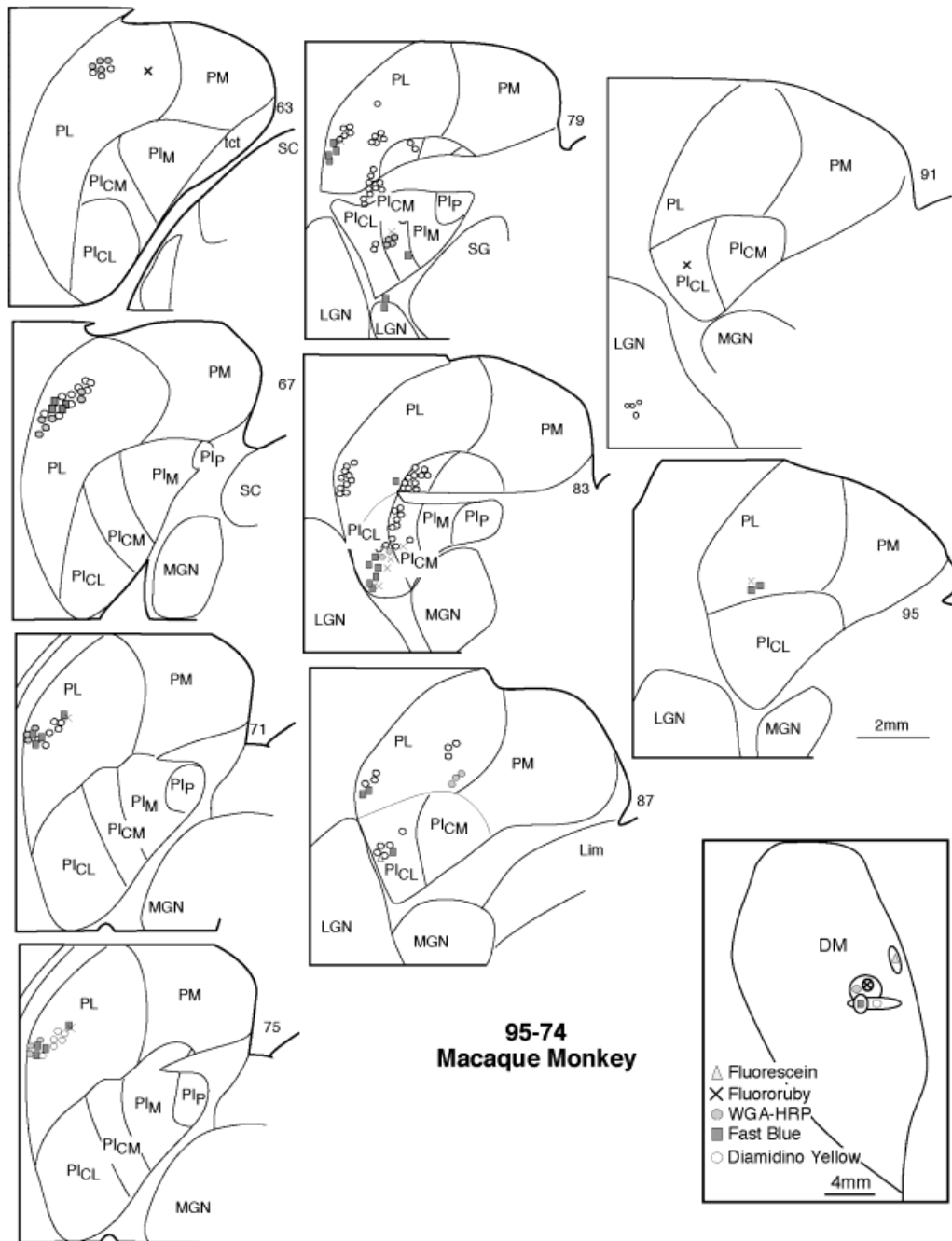


Fig. 11. Injections of five tracers and resulting label in macaque monkey case 95-74. tct, Tractus corticotectalis. For conventions, see Figure 6.

monkeys, and Old World macaque monkeys. In these primates, the presumptive DM interconnects strongly with lateral and inferior divisions of the pulvinar but not with the medial division. Fewer connections were with the LGN, the limitans nucleus, the reticular nucleus, and the claustrum.

In monkeys, in which the application of several histochemical procedures helped to reveal nuclei of the inferior pulvinar, further similarities in connections were observed. In owl monkeys, the inferior pulvinar was de-

scribed initially with posterior (IPp), medial (IPm), and central (IPC) nuclei (Lin and Kaas, 1979, 1980) on the bases of differences in myeloarchitecture and connections with visual cortex and the superior colliculus. More recently, it has become obvious that IPC in monkeys contains two quite different histochemical regions (Cusick et al., 1993; Gutierrez et al., 1995; Stepniewska and Kaas, 1997). The smaller medial part or nucleus, which we call PI_{CM} after Stepniewska and Kaas (1997), stains intensely for CB, whereas the lateral division, PI_{CL} , stains moderately

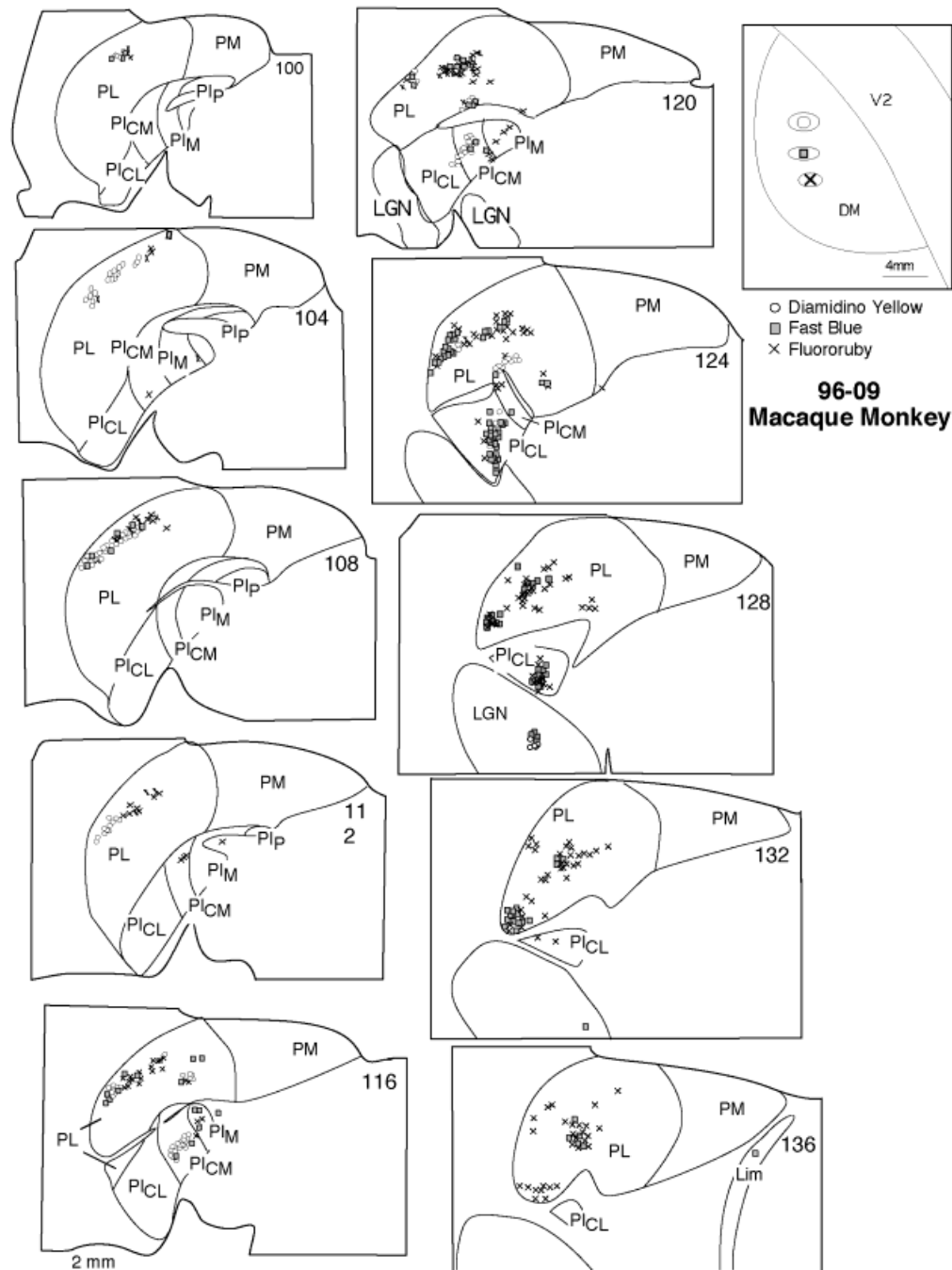


Fig. 12. Illustrations of three injections and resulting label in macaque monkey case 96-09. For conventions, see Figure 3.

for CB, CO, and AChE. Although there is agreement about the validity of these two newly defined subdivisions of the inferior pulvinar, there are differences in interpretation over their full extents, and somewhat different terminologies have been used (Cusick et al., 1993; Gutierrez et al., 1995; Gutierrez and Cusick, 1997; Stepniewska and Kaas, 1997). Our use of histochemical procedures to subdivide the inferior pulvinar allowed us to compare the pulvinar connections of DM in more detail across primate species

(Fig. 14). In all studied groups of monkeys, PI_{CM} and PI_{CL} had dense connections with DM, but connections with PI_M and PI_P were sparse or absent. Similar patterns could exist in galagos, in which separate clumps of labeled neurons in the inferior pulvinar were observed, but the histochemical procedures did not reveal subdivisions of the inferior pulvinar. In the lateral pulvinar, connections were also dense in all primates after DM injections, and the important finding here was that separate foci of label were

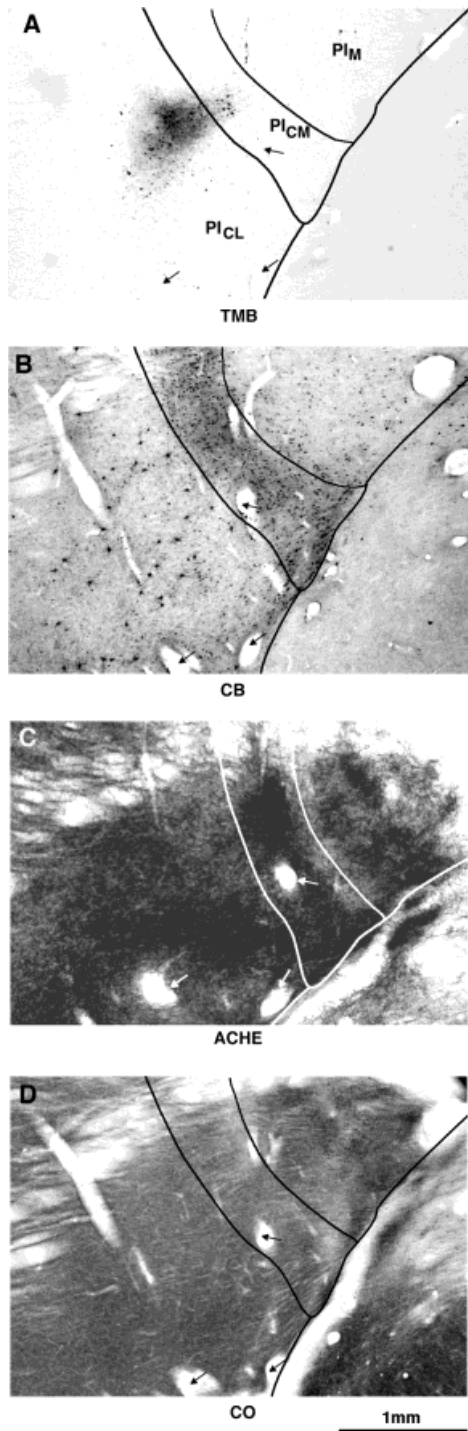


Fig. 13. The location of neurons and terminals labeled with a wheat germ agglutinin conjugated to horseradish peroxidase (WGA-HRP) injection into dorsomedial visual area (DM) relative to chemoarchitectonic divisions of the inferior pulvinar in macaque monkey 94–81. Labeled cells and terminals are located in PI_{CL} and PI_{CM} (A). Arrows indicate blood vessels that are present in adjoining sections stained for tetramethylbenzidine (TMB; A) calbindin (CB; B), acetylcholinesterase (AChE; C), or cytochrome oxidase (CO; D). Lines indicate borders of PI_{CL} , PI_{CM} , and PI_M .

suggestive of two nuclei within the lateral pulvinar, a possibility that has other support (for review, see Kaas and Huerta, 1988).

Our observations on pulvinar connections are consistent with the hypothesis that a homologous visual area exists along the dorsomedial rostral border of V2 in all of these primates. There is additional evidence from microelectrode mapping studies (Krubitzer and Kaas, 1993; Rosa and Schmid, 1995; Rosa et al., 1997) and studies of topographic patterns of cortical connections (Krubitzer and Kaas, 1990, 1993; Beck and Kaas, 1995, 1996a; Stepniewska and Kaas, 1997) that the moderately myelinated wedge of cortex we define as DM contains a systematic representation of both upper and lower quadrants of the contralateral field and has inputs from both V1 and V2 in prosimians, New World monkeys, and Old World monkeys, as described originally for DM of owl monkeys (Allman and Kaas, 1975; Wagor et al., 1975; Lin and Kaas, 1979, 1980). Together, these results argue for a consistent organization of extrastriate visual cortex in the region of DM across primate taxa. More importantly, the findings are more compatible with the concept of a single, complete visual area, DM, than the proposed alternative that the region consists of a dorsal V3 plus part or all of an adjoining V3A (for reviews of current theories, see Kaas 1996, 1997).

Although the results are compatible with the proposal that DM exists in a range of primate species, we do not know how distinct the pattern of pulvinar connections of the DM region is from those of other visual areas. Because adjoining visual areas tend to have similar cortical connections (Young et al., 1995), similar pulvinar connections are also expected for adjoining and nearby visual areas. Both the inferior and lateral pulvinar have been shown to be connected with a number of visual areas in owl monkeys (Graham et al., 1979), squirrel monkeys (Cusick et al., 1993), marmosets (Dick et al., 1991), and macaque monkeys (Boussaoud et al., 1992). However, clear differences in pulvinar connections have been reported for different cortical areas. For example, we have long known that PI_M is densely interconnected with MT, whereas PI_C (PI_{CM} and PI_{CL} in the present study) is densely interconnected with cortex caudal to MT (Lin and Kaas, 1979, 1980). Given the existence of newly discovered subdivisions of the inferior pulvinar complex, further study could reveal further differences in connection patterns. However, we expect that adjacent areas in the same processing stream (Ungerleider and Mishkin, 1982) will have similar pulvinar connections and that areas at different levels of processing and in different streams will differ more in thalamic connections. DM seems to have mixed dorsal and ventral stream cortical inputs, but its major outputs to posterior parietal cortex identify DM more strongly with the dorsal stream. However, DM has quite different inferior pulvinar connections than MT. Whereas MT has dense IP_M inputs (Lin and Kaas, 1980; Wall et al., 1982; Maunsell and Van Essen, 1983; Standage and Benevento, 1983; Ungerleider et al., 1984; Dick et al., 1991; Cusick et al., 1993), DM injections labeled very few neurons in PI_M . Instead, both V2 and caudal DL (V4) appear to have dense connections with PI_{CM} and PI_{CL} in common with DM (Lin and Kaas, 1979; Cusick et al., 1993; Stepniewska and Kaas, 1995).

Another important finding of the present study was the clear evidence that small numbers of neurons in the LGN of all studied primates project to DM. The LGN of cats has

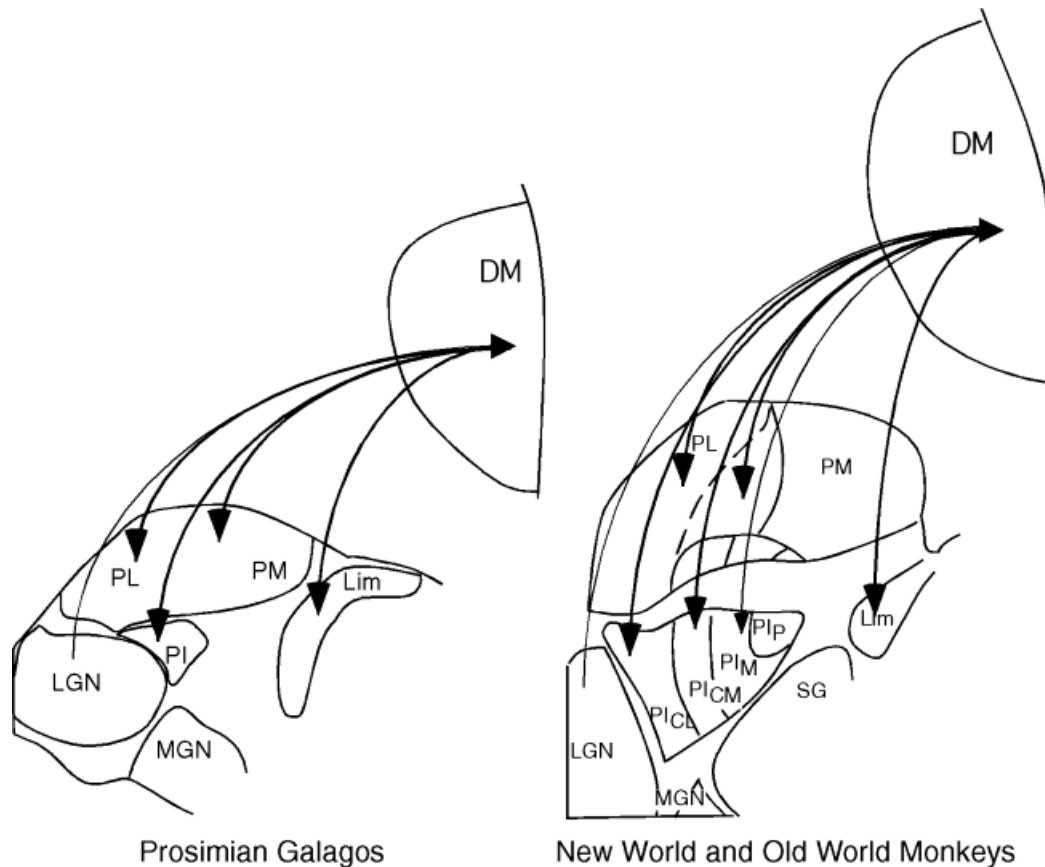


Fig. 14. Summary of subcortical connections with the dorsomedial visual area. In prosimian galagos, subdivisions of the pulvinar are not obvious in architectural stains, and evidence from connections did not delimit divisions of the inferior pulvinar. Labeled neurons located medially may have been located in medial pulvinar (PM) or lateral pulvinar (PL), but architecture was not well differentiated, and

borders were unclear. In galagos and monkeys, dense connections are shown with thick arrows, and sparse connections are shown with thin arrows. Reciprocal connections are shown with double arrowheads, and projections that are not reciprocal are shown with a single arrowhead. The location of arrows in DM has no significance and does not indicate a topographic pattern of thalamic connections.

long been known to project widely to extrastriate cortex (for review, see Stone, 1983), but the severe retrograde degeneration and loss of neurons in the LGN after V1 lesions in primates (Weller et al., 1984) and the early use of less sensitive tracers (Wilson and Cragg, 1967; Garey and Powell, 1971; Hendrickson et al., 1978) led to the early assumption that LGN neurons project exclusively to V1. However, sensitive retrograde transport methods subsequently revealed that a few LGN neurons in monkeys do project to extrastriate cortex (Wong-Riley, 1976; Yukie and Iwai, 1981). However, it is not yet certain how many extrastriate areas receive LGN projections or whether this extrastriate projection pattern is species variable. In macaque monkeys, LGN neurons have been shown to project to V2 (Bullier and Kennedy, 1983; Stepniewska and Kaas, 1995), V4 (Lysakowski et al., 1988; Cowey and Stoerig, 1989), and even inferotemporal cortex (Hernandez-Gonzalez et al., 1994). These projecting neurons are always few in number, and they vary in laminar location. Yet, they could be important in mediating aspects of vision in extrastriate cortex after lesions of V1 (Cowey and Stoerig, 1989). The present results demonstrate that DM receives some direct LGN input in prosimians, New World monkeys, and Old World monkeys, demonstrating a type of phylogenetic consistency across these primates. The ten-

dency for this projection to originate from neurons in the magnocellular layers, especially in monkeys, is consistent with the premise that DM is more involved in the dorsal stream than in the ventral stream of processing.

The present results also revealed that DM has connections with the limitans nucleus, the claustrum, and the superior colliculus. Such connections have been described before for extrastriate visual areas in primates (Spatz and Tigges, 1973; Spatz, 1975; Norden et al., 1978; Riche and Lanoir, 1978; Graham et al., 1979; Raczkowski and Diamond, 1980; Tigges and Tigges, 1981; Tigges et al., 1982; Wall et al., 1982; Fries, 1984; Boussaoud et al., 1992), and such connections may exist for all or most visual areas. In general, more caudal visual areas project more superficially in the superior colliculus than more rostral visual areas (Graham et al., 1979; for review, see Kaas and Huerta, 1988), and DM terminations were in the middle of this termination range across these primates, as described previously for owl monkeys (Graham et al., 1979). Here, for the first time, we describe projections from DM to the intermediate layers in addition to the superficial layers of the superior colliculus. MT has also been reported as projecting to the superficial and intermediate layers in marmosets (Spatz, 1975). These results place DM with MT as intermediate to caudal areas (e.g., areas 17 and 18) that

project only to superficial layers (Graham et al., 1979; Tigges and Tigges, 1981; Wall et al., 1982; Fries, 1984) and rostral areas (e.g., VPP, inferotemporal cortex and the frontal eye field) that project to intermediate and deep layers of the superior colliculus (Kunzle et al., 1976; Kunzle, 1977; Fries, 1984).

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